

- 5 An isolated spherical conductor has charge q , as shown in Fig 5.1.

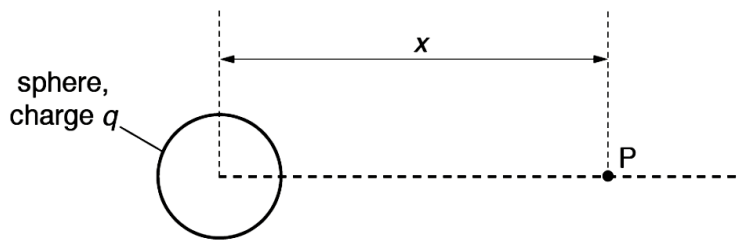


Fig. 5.1

Point P is a movable point with a distance of x from the centre of the sphere. The variation with distance x of the electric potential V at a point P due to the charges on the sphere is shown in Fig. 5.2.

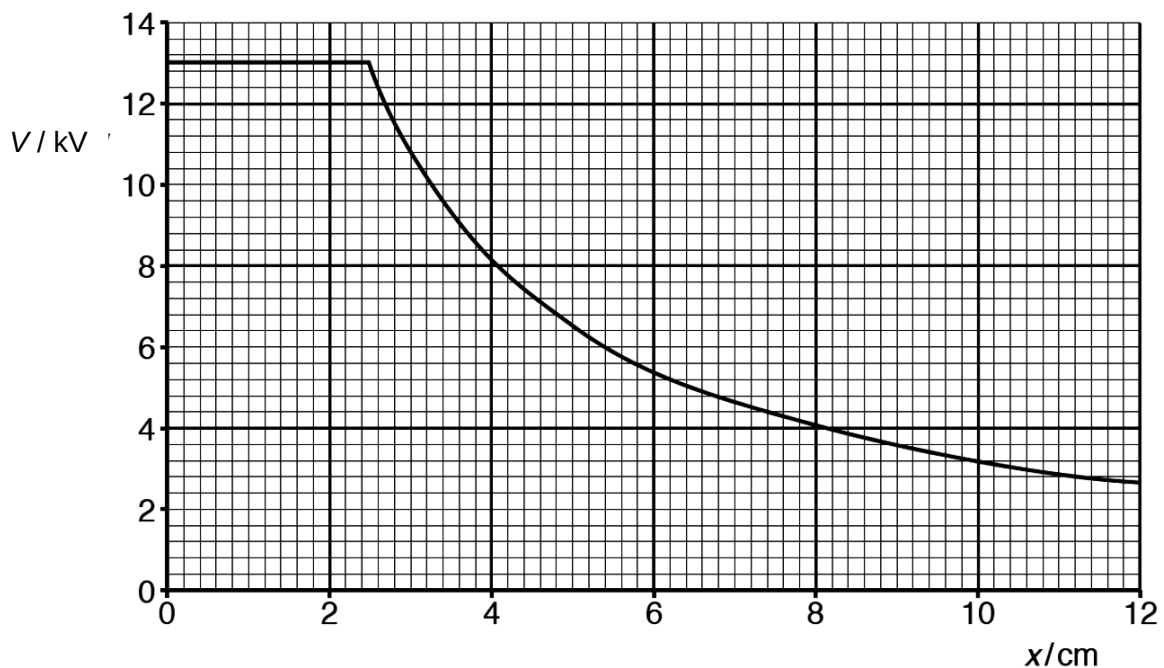


Fig. 5.2

- (a) By making reference to the electric field, explain why the potential is constant for $x = 0$ cm to $x = 2.5$ cm.

- For $x = 0$ cm to $x = 2.5$ cm is the region inside the spherical conductor, so no electric field / electric field strength is zero inside [B1].
- Since electric field strength is equal to the negative of potential gradient (i.e. $E = -\frac{dV}{dx}$), potential gradient is zero, therefore potential must remain constant [B1]. [2]

(b) Use Fig. 5.2 to determine the acceleration of a proton at point P where $x = 5.0$ cm.

Method 1:

- Tangent drawn at $x = 5.0$ cm and chosen gradient coordinates must be longer than half the tangent line, to find E. [C1]
- Use of N2L and $a = \frac{qE}{m}$ seen. [C1]
- Correct calculation of final ans. [A1]

Method 2:

- Determine charge, Q, of sphere using formula for isolated point charge $V = \frac{Q}{4\pi\epsilon_0 r}$. [C1]
- Determine electric force using Coulomb's law $F = \frac{Qq}{4\pi\epsilon_0 r^2}$ and N2L to find a. [C1]
- Correct calculation of final ans. [A1]

acceleration = **1.2 to 1.4×10^{13}** m s⁻² [3]

(c) Describe and explain the variation of the speed of the proton when it moves from $x = 5.0$ cm to $x = 9.0$ cm.

Method 1 (COE approach):

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- Electric potential decreases, electric potential energy decreases, kinetic energy increases [C1]
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- So speed increases [A1].
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Method 2 (N2L approach):

-[2]
- Electric field strength decreases, electric force decreases [C1]
 - Acceleration decreases, so speed increases at a decreasing rate [A1]

(d) If the proton has a speed of 1.3×10^5 m s⁻¹ initially at $x = 5.0$ cm, calculate the speed of the proton when it is at $x = 9.0$ cm.

$$Ep_i + Ek_i = Ep_f + Ek_f \text{ [C1]}$$

$$(1.6 \times 10^{-19})(6.4 \times 10^3) + \frac{1}{2}(1.67 \times 10^{-27})(1.3 \times 10^5)^2 = (1.6 \times 10^{-19})(3.6 \times 10^3) + \frac{1}{2}(1.67 \times 10^{-27})v_f^2 \text{ [C1]}$$

$$v_f = 7.44 \times 10^5 \text{ m s}^{-1} \text{ [A1]}$$

speed =m s⁻¹ [3]

[Total: 10]

6 (a) Two cylindrical resistors M and N of the same material are connected parallel in Fig. 6.1